COMSATS University Islamabad

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Department Of Computer Science

Course	*Information Security*
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Lab Assignment 1

Caesar Cipher Implementation

➢ Introduction

The Caesar Cipher is one of the oldest and simplest encryption techniques. It is a type of substitution cipher in which each letter in the plaintext is shifted by a fixed number of positions in the alphabet.
For example, if the shift value is 3:

- A becomes D
- B becomes E
- Z becomes C (wrap around)

This lab assignment implements encryption, decryption using Python.

➢ Program Explanation (Line-by-Line)

1. Function Definition for Encryption

```
# Function to encrypt the text
def caesar_encrypt(text, shift):
```

This function takes two inputs:
text → the original message
shift → number of positions to shift letters

2. Creating an Empty String

```
encrypted_text = ""  # Empty string to store result
```

This variable stores the encrypted result.

3. Loop Through Each Character

```
for char in text:
```

This loop processes each character one by one.

4. Checking Uppercase Letters

```
# Check if character is uppercase letter
if char.isupper():
```

If the character is uppercase:

- Convert to ASCII using ord()
- Apply shift
- Use % 26 to wrap around alphabet
- Convert back using chr()

5. Checking Lowercase Letters

```
# Check if character is lowercase letter
elif char.islower():
```

Same logic but starting ASCII is different (97 instead of 65).

6. Handling Special Characters

```
# If not a letter (space, number, symbol), keep it same
else:
    encrypted_text += char
```

Spaces, numbers, and symbols remain unchanged.

7. Decryption Function

```
# Function to decrypt the text
def caesar_decrypt(ciphertext, shift):
    decrypted_text = ""
```

Decryption works the same way but subtracts the shift instead of adding it.

8. Main Program

The user:

- Enters message
- Enters shift value
- Program prints encrypted text
- Program prints decrypted text

➢ Example Execution

Input:

Message: Hello World!

Shift: 3

Output:

Encrypted: Khoor Zruog!

Decrypted: Hello World!

➢ Security Analysis

Strengths

- Very simple and easy to understand.
- Introduces fundamental cryptography concepts.
- Useful for learning substitution ciphers.

Weaknesses

- Only 25 possible keys.
- Easily breakable using brute force.
- Vulnerable to frequency analysis attacks.
- Not secure for real-world communication.

Security Conclusion

The Caesar Cipher is not secure by modern standards. It is only suitable for educational purposes and understanding basic encryption principles.

➢ Conclusion

This lab assignment successfully demonstrates encryption and decryption using the Caesar Cipher. The implementation maintains uppercase and lowercase letters and keeps spaces and special characters unchanged. Through this exercise, we understand both the working mechanism and security limitations of classical encryption techniques.